CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of MSc Physics Examination March 2018

PS712 Classical Mechanics-I

Date: 27/03/2018 Day: Tuesday Time: 10:00 AM To 10:30 AM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. **10**

1. Constraints that depend upon velocities are called as

(a) holonomic

(b) non-holonomic

(c) scleronomic-holonomic

(d) rheonomic-holonomic

2. The constraint of a particle moving on a straight line along x-axis is

(a) Holonomic

(b) Non-holonomic

(c) Scleronomic-holonomic

(d) Rheonomic-holonomic

3. Hamiltonian of a system is

(a) the total energy of the system

(b) independent of generalized coordinates of the system

(c) independent of time

(d) none of the above

4. The number of generalized coordinates for particle moving on a circle of radius r is

(a) One

(b) Two

(c) Three

(d) Anything between one and two

5. Generalized momentum corresponding to a generalized coordinate q_i is defined as

(a) $\frac{\partial L}{\partial \dot{q}_i}$

(b) $\frac{\partial L}{\partial q_i}$

(c) $\frac{\partial L}{\partial p_i}$

(d) $q_i \dot{q}_i$

6. The Hamiltonian can be constructed from the Lagrangian using the formula

(a) $H = \dot{p}_i \dot{q}_i - L$

(b) $H = \sum p_i \dot{q}_i - L$

(c) $H = \sum \dot{p}_i \dot{q}_i - L$

(d) $H = 1/L$

7. In Kepler's problem, the shape of the orbit of a planetary motion will be circular if the energy E is such that

(a) $E < 0$

(b) $E > 0$

(c) $V_{min} < E < 0$

(d) $E = V_{min}$

8. The Hamilton's equations of motion are given by

(a) $\dot{p}_i = -\frac{\partial H}{\partial q_i};\ \dot{q}_i = \frac{\partial H}{\partial p_i}$

(b) $\dot{p}_i = \frac{\partial H}{\partial q_i};\ \dot{q}_i = -\frac{\partial H}{\partial p_i}$

(c) $\dot{p}_i = -\frac{\partial H}{\partial q_i};\ \dot{q}_i = -\frac{\partial H}{\partial p_i}$

(d) $\dot{p}_i = \frac{\partial H}{\partial q_i};\ \dot{q}_i = \frac{\partial H}{\partial p_i}$

9. If the Lagrangian is cyclic in q_j, then

(a) p_j is conserved

(b) q_j appears in the Lagrangian

(c) p_j is not conserved

(d) $\dot{q}_j$ does not appear in the Lagrangian.

10. The Lagrangian of a particle of mass m having one dimensional simple harmonic motion with frequency ω is given by

(a) $\frac{m}{2}(\dot{x}^2 - \omega^2 x^2)$

(b) $\frac{m}{2}(\dot{x}^2 + \omega^2 x^2)$

(c) $\frac{m}{2}(\dot{x}^2 - \omega^2 \dot{x}^2)$

(d) $\frac{m}{2}(x^2 - \omega^2 x^2)$

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of MSc Physics Examination March 2018

PS712 Classical Mechanics-I

Date: 27/03/2018 Day: Tuesday Time: 10:30 AM To 1:00 PM Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

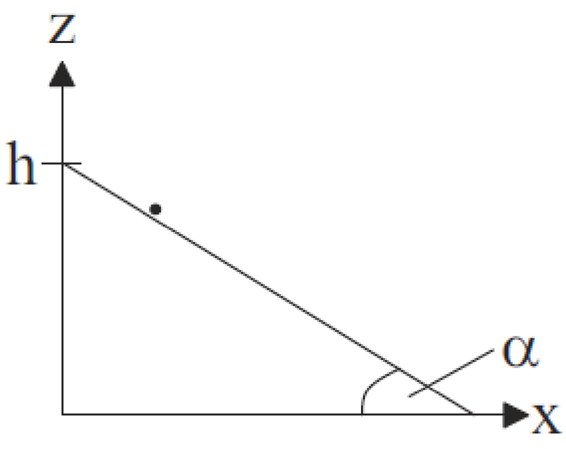
Q - II Answer the following questions as directed **10**

1. Write down the principle of conservation of momentum and energy for a system of particles. **2**

2. Write down the Hamiltonian of a particle of mass m undergoing simple harmonic motion. Draw the phase space diagram. **2**

3. Write down the Lagrangian of particle of mass m moving freely in space using Cartesian, spherical and cylindrical coordinates. **3**

4. Write down the constrain of particle moving on an inclined plane (without friction) as shown in the figure below. Obtain the number of degrees of freedom for the particle. Classify the constraint of the particle. **3**



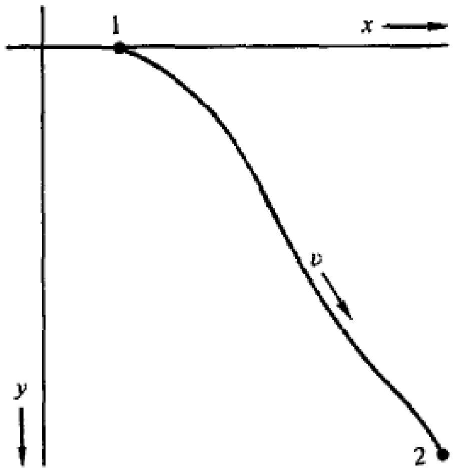
SECTION – II

Q-III Answer the following questions as directed **15**

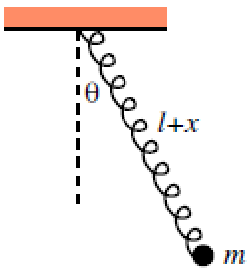
1. Using calculus of variation find the shortest distance between two points in a plane. **5**

OR

The brachistochrone problem: Find the curve joining two points, along which a particle falling from rest under the influence of gravity travels from the higher to the lower point in the least time.

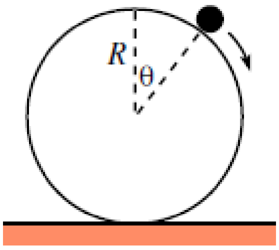


2. Consider a simple pendulum made out of a spring with a mass m on the end as shown in the figure below. The spring is arranged to lie in a straight line (this can be achieved by wrapping the spring around a massless rigid rod). The equilibrium length of the spring is l. Let the spring has length $l + x(t)$, and let its angle with the vertical be $\theta(t)$. Find the Lagrangian of this system and hence obtain the equations of motions for x and θ. **5**

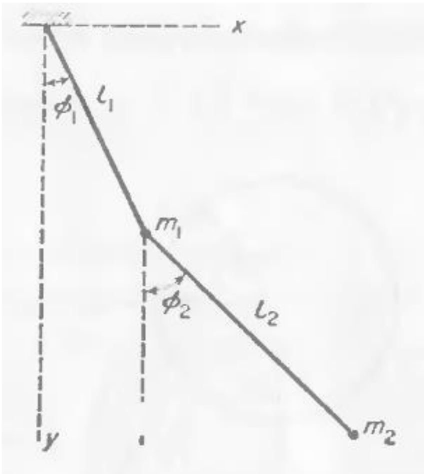


OR

Find the Lagrangian and hence the equations of motions for a particle of mass m sliding on the frictionless surface of a sphere of radius R.



3. Find the Hamiltonian of a coplanar double pendulum as shown in the figure below. Then obtain the Hamilton equations of motion. **5**



OR

Find the Hamiltonian and Hamilton equations of motion of a simple pendulum of mass m_2, with a mass m_1 at the point of support which can move on a horizontal line lying in the plane in which m_2 moves.